Viper - Using MISP from your terminal

MISP - Intelligence Sharing

CIRCL / Team MISP Project

MISP Project

https://www.misp-project.org/

CRI MISP 2024



Viper - Main ideas

Viper is a **binary analysis and management framework**. *Its fundamental objective is to provide a solution to* **easily organize** *your collection of* **malware** *and* **exploit samples** *as well as your collection of* **scripts** *you created or found over the time to facilitate your daily research. Think of it as a* **Metasploit for malware researchers**: *it provides a terminal interface that you can use to* **store**, **search** *and* **analyze** *arbitrary files with and a framework to* **easily create plugins** *of any sort.*

Viper - Main ideas

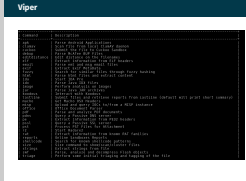
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- **Solid CLI**
- Plenty of modules (PE files, *office, ELF, APK, ...)
- Connection to **3rd party services** (MISP, VirusTotal, cuckoo)
- Connectors to **3rd party tools** (IDA, radare)
- **Locale storage** of your own zoo
- Django interface is available (I've been told)

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Viper

```
+--------------+-----------------------------------------------------------------------------------------+
| Command      | Description                                                                             |
+--------------+-----------------------------------------------------------------------------------------+
| apk          | Parse Android Applications                                                              |
| clamav       | Scan file from local ClamAV daemon                                                      |
| cuckoo       | Submit the file to Cuckoo Sandbox                                                       |
| debup        | Parse McAfee BUP Files                                                                  |
| editdistance | Edit distance on the filenames                                                          |
| elf          | Extract information from ELF headers                                                    |
| email        | Parse eml and msg email files                                                           |
| exif         | Extract Exif MetaData                                                                   |
| fuzzy        | Search for similar files through fuzzy hashing                                          |
| html         | Parse html files and extract content                                                    |
| ida          | Start IDA Pro                                                                           |
| idx          | Parse Java IDX files                                                                    |
| image        | Perform analysis on images                                                              |
| jar          | Parse Java JAR archives                                                                 |
| koodous      | Interact with Koodous                                                                   |
| lastline     | Submit files and retrieve reports from LastLine (default will print short summary)      |
| macho        | Get Macho OSX Headers                                                                   |
| misp         | Upload and query IOCs to/from a MISP instance                                           |
| office       | Office Document Parser                                                                  |
| pdf          | Parse and analyze PDF documents                                                         |
| pdns         | Query a Passive DNS server                                                              |
| pe           | Extract information from PE32 headers                                                   |
| pssl         | Query a Passive SSL server                                                              |
| pst          | Process PST Files for Attachment                                                        |
| r2           | Start Radare2                                                                           |
| rat          | Extract information from known RAT families                                             |
| reports      | Online Sandboxes Reports                                                                |
| shellcode    | Search for known shellcode patterns                                                     |
| size         | Size command to show/scan/cluster files                                                 |
| strings      | Extract strings from file                                                               |
| swf          | Parse, analyze and decompress Flash objects                                             |
| triage       | Perform some initial triaging and tagging of the file                                   |
```

- Full featured **CLI for MISP**
- **Remote storage** of your zoo
- Search / **Cross check with VirusTotal**
- Create / Update / Show / Publish Event
- Download / Upload Samples
- Mass export / Upload / Download
- Get Yara rules

PyMISP & Viper
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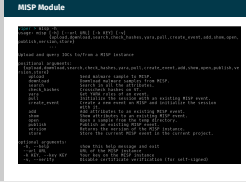
MISP Module

```
viper > misp -h
usage: misp [-h] [--url URL] [-k KEY] [-v]
            {upload,download,search,check_hashes,yara,pull,create_event,add,show,open,
publish,version,store}
            ...

Upload and query IOCs to/from a MISP instance

positional arguments:
  {upload,download,search,check_hashes,yara,pull,create_event,add,show,open,publish,ve
rsion,store}
    upload              Send malware sample to MISP.
    download            Download malware samples from MISP.
    search              Search in all the attributes.
    check_hashes        Crosscheck hashes on VT.
    yara                Get YARA rules of an event.
    pull                Initialize the session with an existing MISP event.
    create_event        Create a new event on MISP and initialize the session
                        with it.
    add                 Add attributes to an existing MISP event.
    show                Show attributes to an existing MISP event.
    open                Open a sample from the temp directory.
    publish             Publish an existing MISP event.
    version             Returns the version of the MISP instance.
    store               Store the current MISP event in the current project.

optional arguments:
  -h, --help            show this help message and exit
  --url URL             URL of the MISP instance
  -k KEY, --key KEY     Your key on the MISP instance
  -v, --verify          Disable certificate verification (for self-signed)
```

Viper & VT

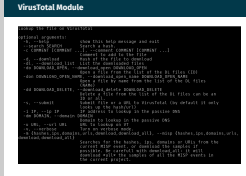
- Searches for hashes/ips/domains/URLs from the current MISP event, or download the samples
- Download samples from current MISP event
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VirusTotal Module

```
Lookup the file on VirusTotal

optional arguments:
  -h, --help            show this help message and exit
  --search SEARCH       Search a hash.
  -c COMMENT [COMMENT ...], --comment COMMENT [COMMENT ...]
                        Comment to add to the file
  -d, --download        Hash of the file to download
  -dl, --download_list  List the downloaded files
  -do DOWNLOAD_OPEN, --download_open DOWNLOAD_OPEN
                        Open a file from the list of the DL files (ID)
  -don DOWNLOAD_OPEN_NAME, --download_open_name DOWNLOAD_OPEN_NAME
                        Open a file by name from the list of the DL files
                        (NAMe)
  -dd DOWNLOAD_DELETE, --download_delete DOWNLOAD_DELETE
                        Delete a file from the list of the DL files can be an
                        ID or all.
  -s, --submit          Submit file or a URL to VirusTotal (by default it only
                        looks up the hash/url)
  -i IP, --ip IP        IP address to lookup in the passive DNS
  -dm DOMAIN, --domain DOMAIN
                        Domain to lookup in the passive DNS
  -u URL, --url URL     URL to lookup on VT
  -v, --verbose         Turn on verbose mode.
  -m {hashes,ips,domains,urls,download,download_all}, --misp {hashes,ips,domains,urls,
download,download_all}
                        Searches for the hashes, ips, domains or URLs from the
                        current MISP event, or download the samples if
                        possible. Be carefull with download_all: it will
                        download *all* the samples of all the MISP events in
                        the current project.
```

Extra features

- Link to a MISP event
- Local storage of the MISP event
- On the fly cross-check of MISP atributes with 3rd party services
- Never leaving your CLI!

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Other modules

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- Can launch Radare2 or IDA

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Passive SSL

```
viper > pssl -h
usage: pssl [-h] [--url URL] [-u USER] [-p PASSWORD] [-i IP] [-c CERT]
            [-f FETCH] [-v] [-m {ips}]

Query a Passive SSL server

optional arguments:
  -h, --help            show this help message and exit
  --url URL             URL of the Passive SSL server (No path)
  -u USER, --user USER  Username on the PSSL instance
  -p PASSWORD, --password PASSWORD
                        Password on the PSSL instance
  -i IP, --ip IP        IP to query (can be a block, max /23).
  -c CERT, --cert CERT  SHA1 of the certificate to search.
  -f FETCH, --fetch FETCH
                        SHA1 of the certificate to fetch.
  -v, --verbose         Turn on verbose mode.
  -m {ips}, --misp {ips}
                        Searches for the ips from the current MISP event
```


Passive SSL



Passive DNS

```
viper > pdns -h
usage: pdns [-h] [--url URL] [-u USER] [-p PASSWORD] [-v] [-m {ips,domains}]
            [query]

Query a Passive DNS server

positional arguments:
  query                 Domain or IP address to query

optional arguments:
  -h, --help            show this help message and exit
  --url URL             URL of the Passive DNS server
  -u USER, --user USER  Username on the PDNS instance
  -p PASSWORD, --password PASSWORD
                        Password on the PDNS instance
  -v, --verbose         Turn on verbose mode.
  -m {ips,domains}, --misp {ips,domains}
                        Searches for the ips or domains from the current MISP
                        event
```


Passive DNS



Q&A

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